

Junior Castillo

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Education

University of California, San Diego (UCSD)

Sep 2023 – Dec 2026

B.S. Mathematics–Computer Science, *Provost Honors*

GPA: 3.7/4.0

- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Computer Organization, Systems Programming, Computer Networking, Deep Learning, Recommender Systems, Graph Theory, Probability, Stochastic Processes, Parallel Computing, Machine Learning

Experience

San Diego Supercomputer Center (SDSC)

Oct 2025 – Present

Software Engineer Intern

La Jolla, CA

- Designed and developed a **Go** backend service that replaced a legacy **Python** pipeline, automating customer and order sync between **Microsoft Dynamics CRM** and **StruxureWare** for the SDSC Billing team.
- Built two reusable **HTTP API clients** in **Go**: a **Dynamics OData v8.0** client with **NTLM authentication** and cookie-jar session reuse, and a Basic Auth **REST** client for StruxureWare, eliminating duplicated request logic.
- Engineered a **multi-source change-detection layer** that unions recently modified accounts, contacts, and colocation orders and deduplicates them, ensuring no customer update is missed between sync cycles.
- Implemented **idempotent upsert workflows** that probe the target system before writes, preventing duplicate records and enabling safe re-runs after partial failures.

Software Developer Intern

June 2025 – Sep 2025

- Designed and maintained the **Firebase** backend that securely managed user profiles, listings, chat messages, and image data for a mobile marketplace app.
- Developed mobile **UI components** and screens for listings, profiles, and chat in **React Native**, accelerating feature delivery across the team.
- Utilized **GitLab** for **version control** and task management across multiple contributors.

Projects

TritonTalk - Reliable Transport Protocol | C, POSIX Sockets, Make, Docker

Apr 2026

- Designed and built a **sliding-window transport protocol** in **C** that delivered messages reliably over a simulated lossy network, recovering from corrupted frames via **CRC-8** and reordering via per-sender receive windows.
- Implemented **cumulative and selective acknowledgements** that reduced redundant retransmissions across concurrent hosts, enabling correct message reassembly under packet loss.
- Engineered **TCP-style congestion control** (slow start, **AIMD**, fast retransmit) that maximized throughput across test scenarios, with per-RTT **cwnd/ssthresh** diagnostics logged to CSV.

Cookie Trailer | React, Konva.js

Sep 2025 – Dec 2025

- Designed and built the full **shopping screen system** using **Konva.js** and **React**, enabling players to select ingredients, view live cost calculations, and track balances in real time.
- Developed reusable **UI components** (purchase buttons, recipe viewers, order receipts, shuffle controls) with hover and click effects, standardizing interaction patterns across the game.
- Engineered a dynamic **customer order system** tracking daily demand and purchases, visually displaying receipts to drive core gameplay.

Taylorred Text - Hackathon Project | JavaScript, Vite, Gemini API

June 2025

- Developed a **Chrome extension** that makes articles interactive for AI-powered language learning, reducing friction of vocabulary lookup while reading.
- Integrated the **Gemini API** to extract key vocabulary and provide translations with example sentences inline with article text.
- Designed a responsive **UI** that displays translations and quizzes without disrupting the reading experience.

Technical Skills

- **Programming:** Python, Java, C, C++, JavaScript, TypeScript, Go
- **Frameworks & Libraries:** React, React Native, Tailwind CSS, Expo Go, PyTorch, Konva, Swagger, Postman
- **Markup & Styling:** LaTeX, HTML, CSS
- **Tools & Software:** Firebase, Git, GitHub, GitLab, Excel, MATLAB, Figma, Docker, OpenCL